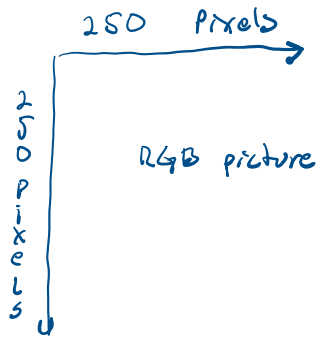


Neural network lecture continued



Dense NN universal function approximator

$250 \times 250 \times 3 = 187,500$ layers input

120 outputs

250+ nodes hidden layer

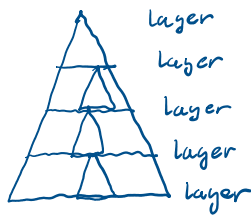
50×10^9 connections, huge amount

CNN - Convolutional NN

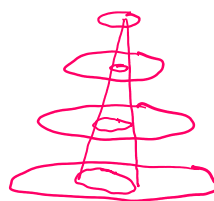
Receptive Fields in Vision

gets to use a division of labor

nodes stacked like a pyramid scheme or a management scheme



can be better in time, and less computational overhead



limited and controlled redundancy unlike

dense systems like this

convolutions in Computer Vision

* Simple box blur convolution
avg effect - produces a slight blur

convolution is a small transformer

low level processing $\rightarrow$ low level features extraction
high " " $\rightarrow$ high " " "

Detecting horizontal lines - only horizontal lines

Edge detection - all lines no matter the orientation

hadamard product (claude.ai, what is this?)

of convolutions are learned, not given

Introducing Receptive Fields: Convolutions

a small filter that is applied at points of the image
and slid over the image
convolution will be a matrix

convolution will produce an activation map
each filter will produce its own activation map
(claude.ai, what is a tensor?)

Dimensions of input/output volumes

stride of 1 (claude.ai, what class did I take that has stride as
a term? I think it was operating systems CS4800)

would need padding maybe

$$32 \times 32 \times 3 \rightarrow 1 \times 28 \times 28$$

Basic CNN

input layer $\rightarrow$... $\rightarrow$ output layer
Feature extraction classification at output, what am I looking at
regression, what we want the CNN to do

other layers:

dropout layer, pooling layer

dropout - Training phase, Testing phase

learn less so we can apply the model
to other datasets

used to prevent overfitting - zero some nodes out

(claude.ai, fill in with more info)

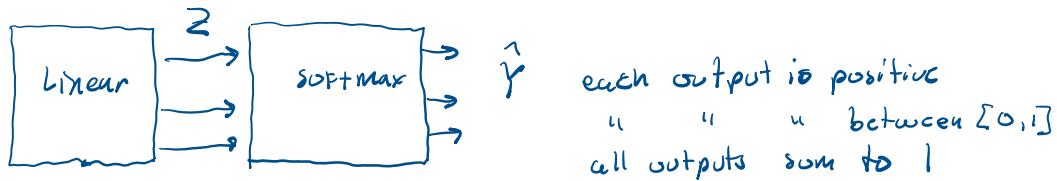
Downsampling - pooling layer

max pooling is commonly used - get the largest value in a small subregion
purpose - reduce the bloat of layers to enhance computation

translation invariance

Soft max layer - used in the activation function in the last convo layer
or classification layer

Squashes them to produce probabilities
normalization, gives values that sum to one



not real probabilities - but we can think of them as probabilities
pseudo-probabilities

makes big numbers bigger, small numbers smaller

makes the winner stand out - no doubt who wins

Layer capture features

LeNet: First CNN Architecture

LeCun Bottou Bengio 1998
IEEE

ImageNet dataset - large dataset - and competitions ILSVRC

images were classified by humans

WordNet lexicon database - 115k unique words - org'd into 176k synsets

AlexNet: ILSVRC 2012 winner Hinton 2012 Toronto Canada

Neuro Net revolution - 2012 paper

ResNet - 2015 winner